Lecture 8 - Continuity and Differentiability

Garrett Credi

10/01/2026

Happy Spooky Season :)

Announcements

1. Exam 1 is next Tuesday (10/06/2026) in class.
2. Exam 1 Review Session is next Monday (10/05/2026) from 6:10-8:10 PM in Vincent Hall 1.
3. All quizzes will be graded by Friday Night. If you want me to look at your homeworks, please email me and I'll grade them specifically.
4. Practice Exam Solutions will be released after class today.
5. Compiled lecture slides also released after class today.

Quiz Time

Quiz Time! Let's get into neat rows and spread out - only then will I pass out the quizzes. DO NOT START OR FLIP OVER THE QUIZ UNTIL I OFFICIALLY START THE QUIZ. If you have any questions about the rules for quizzes, refer to the slide in Lecture 2 or ask me directly. No cheating as per usual.

Continuity

Continuity Recap I

Recall our definitions of continuity for a function $f(x)$:

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Continuity Recap II

Also recall that: if $f(x)$ and $g(x)$ are continuous at $x = a$ and $h(x)$ is continuous at $f(a)$ then:

- $f(x) + g(x)$ is continuous at $x = a$
- $f(x) - g(x)$ is continuous at $x = a$
- $f(x) \cdot g(x)$ is continuous at $x = a$
- If $g(a) \neq 0$, then $\dfrac{f(x)}{g(x)}$ is continuous at $x = a$
- $h(f(x))$ is continuous at $x = a$.

Continuity Recap III

Finally we determined a large family of continuous functions:

- Constant functions $f_c(x) = c$
- Identity function $f(x) = x$
- Polynomials
  $p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$
- Root functions $f(x) = \sqrt[n]{x}$
- Rational functions $\frac{p(x)}{q(x)}$
- Trigonometric functions
  $\sin(x), \cos(x), \tan(x), \arcsin(x)$ etc.
- Exponential functions $f(x) = c^x$
- Logarithmic functions $f(x) = \log_c(x)$

Classifying Failures of Continuity: Jump Discontinuities

What does discontinuity look like?

Jump Discontinuity

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1. The function limits towards a value from the left:

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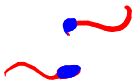
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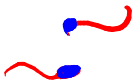
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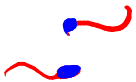
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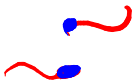
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3. Those values do not agree:

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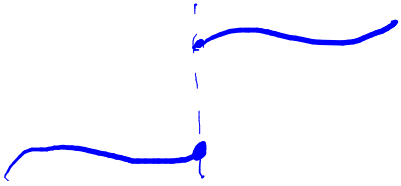
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3. Those values do not agree:

$$\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x) \tag{1}$$

Classifying Failures of Continuity: Jump Discontinuities Visually

Visually, jump discontinuities look like:



→ jump discont

$x = 2$

$x = 0$

$$f(x) = \begin{cases} x+3, & x \le 0 \\ x^2+2, & x > 0 \end{cases}$$

$f(0) = 3$

$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} x+3 = 3$

$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} x^2+2 = 2$

$3 \neq 2$

Classifying Failures of Continuity: Removable Discontinuities

Best kind of discontinuity. "The discontinuity is fixable"

Removable Discontinuity

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Best kind of discontinuity. "The discount is fake"

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2. $f(a)$ either does not exist (we specifically call this a *hole discontinuity* or just a *hole*) or is not equal to $\lim_{x \to a} f(x)$.

Can redefine $f(x)$ *at a single point* to make it continuous at $x = a$.

$$g(x) = \begin{cases} \lim_{x \to a} g(x), & x = a \\ g(x), & x \neq a \end{cases}$$

Removable Discontinuities Visually

Visually,



hole



removable, not hole

Ex. limit definition of derivative

$$\frac{f(x+h)-f(x)}{h}$$

$$f(x)=\begin{cases} x^2, & x\neq 0 \\ 4, & x=0 \end{cases}$$

$$\lim_{x\to 0} f(x)=\lim_{x\to 0} x^2 = 0 \neq f(0)=4$$

$$\bar{f}(x)=\begin{cases} x^2, & x\neq 0 \\ 0, & x=0 \end{cases}$$

Classifying Failures of Continuity: Infinite Discontinuities

Vertical asymptotes



Infinite Discontinuity

A function $f(x)$ has an *infinite discontinuity* if, equivalently:

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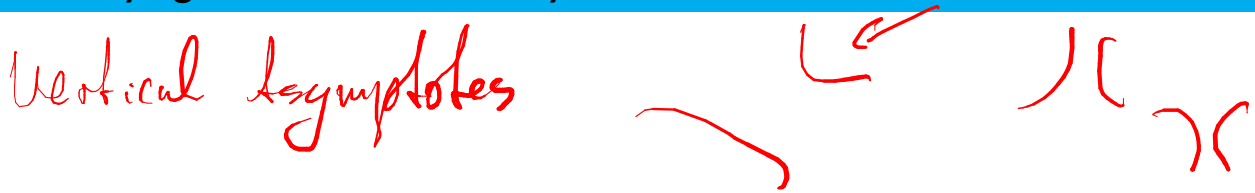


Infinite Discontinuity

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A function $f(x)$ has an *infinite discontinuity* if, equivalently:

- $f(x)$ has a vertical asymptote at $x = a$ or
- $\lim_{x \to a} |f(x)| = +\infty$

Infinite Discontinuities Visually

Visually,



$\lim_{x \to 0^-} 1/x^3 = \frac{1}{\text{neg small}} = \text{neg large}$

$= -\infty$

$\lim_{x \to 0^+} 1/x^3 = +\infty$

$1/x^2$

$\lim_{x \to 0} 1/x^2 = \frac{1}{\text{small pos}} \rightarrow \text{large pos.}$

$\lim_{x \to 0} 1/x^2 = +\infty$

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The *WORST* kinds of discontinuities.

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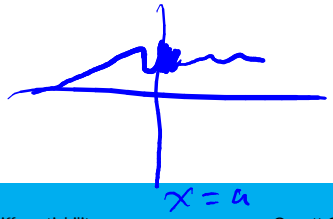


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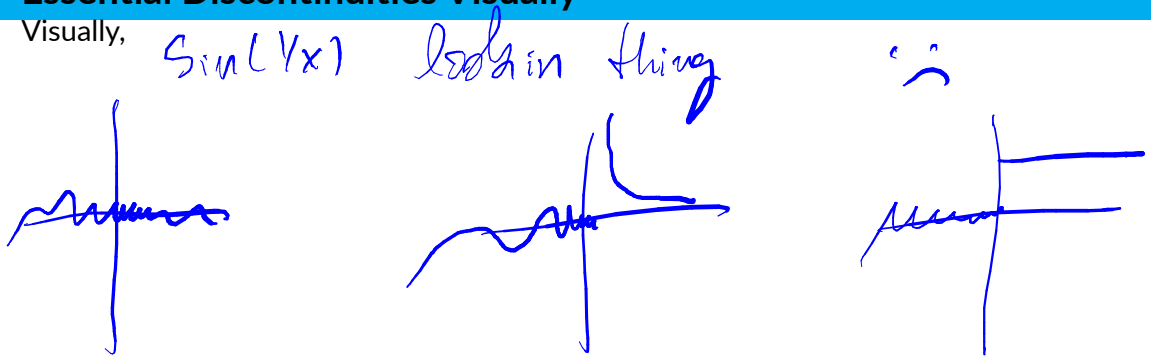
- At least one of the one-sided limits do not exist (not even diverging to $\pm infty$) or
- As $x \to a$, $f(x)$ takes on every number in some interval (v_{min}, v_{max}) *infinitely many times*



$Sin(1/x)$

Essential Discontinuities Visually

Visually,



Some 1/x behavior in a trig func.

sin(1/x) cos(1/x²) :)

What does Continuity buy us?

We already saw one important use of continuous functions:

Direct Substitution

Intermediate Value Theorem

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Hard ⟹ easy sub.

Direct Substitution

If $f(x)$ is continuous at $x = a$ then:

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Continuous functions also have another very important property:

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Tagline: all intermediate output values must be hit by intermediate input values.

IVT Example: Root finding I

Question: For the function

$$f(x) = x^3 - 3x^2 + 2x + 1, \tag{3}$$

which of the following intervals must contain a root of $f(x)$?

1. $[-2, -1]$
2. $[-1, 0]$
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$x^3 - 3x^2 + 2x + 1 = 0$

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Approach 2: IVT.

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Goal: find which of the intervals

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$$f(x) = x^3 - 3x^2 + 2x + 1 \tag{4}$$

switch signs at the endpoints.

IVT Example: Root Finding II

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4. $[1, 2]$

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switch signs at the endpoints.

1. $f(-2) = -8 - 3 \cdot 4 - 2 \cdot 2 + 1 =$
   $-8 - 12 - 4 + 1 = -23$

IVT Example: Root Finding II

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1. $f(-2) = -8 - 3 \cdot 4 - 2 \cdot 2 + 1 = -8 - 12 - 4 + 1 = -23$
2. $f(-1) = -1 - 3 - 2 + 1 = -5$

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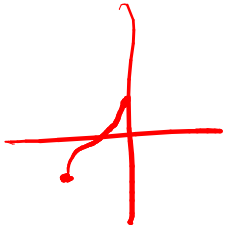
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2. $f(-1) = -1 - 3 - 2 + 1 = -5$
3. $f(0) = 1$

$-5 \leq 0 \leq 1$

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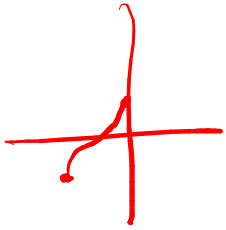
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IVT Example: Root Finding II

Goal: find which of the intervals

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3. $[0, 1]$
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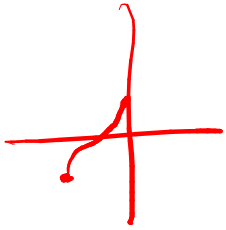
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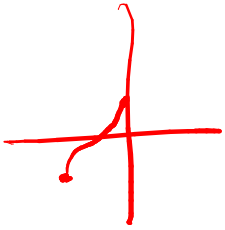
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Only know that a root must be inside $[-1, 0]$.

Differentiability =

Linearly Approx

Differentiability Recap

Again, recall that:

Differentiability Recap

For every $a < c < b$, $f(x)$ diff'able at $x = c$

Again, recall that:

1. $f(x)$ is differentiable at $x = a$: $\lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$
2. $f(x)$ is differentiable on (a, b):
3. $f(x)$ is differentiable everywhere:
4. $f(x)$ is differentiable

SAME as Cont.

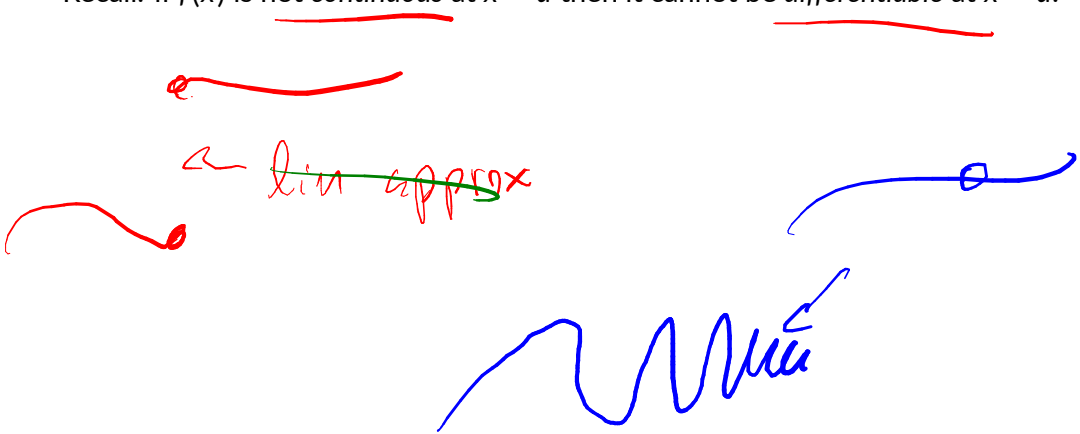
Classifying Failures of Differentiability: Discontinuities

We can already classify some behavior making functions differentiable.

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Recall: if $f(x)$ is not *continuous* at $x = a$ then it cannot be *differentiable* at $x = a$.



Classifying Failures of Differentiability: Corners

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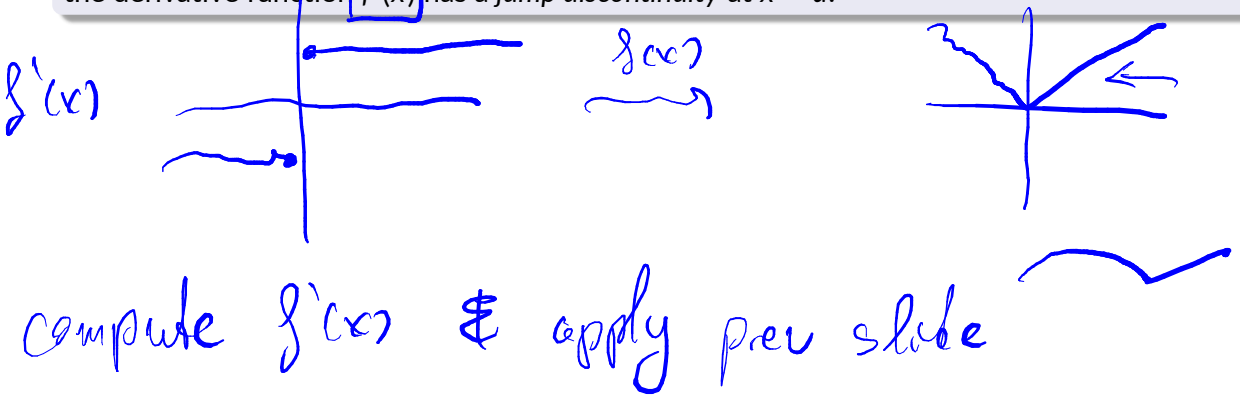
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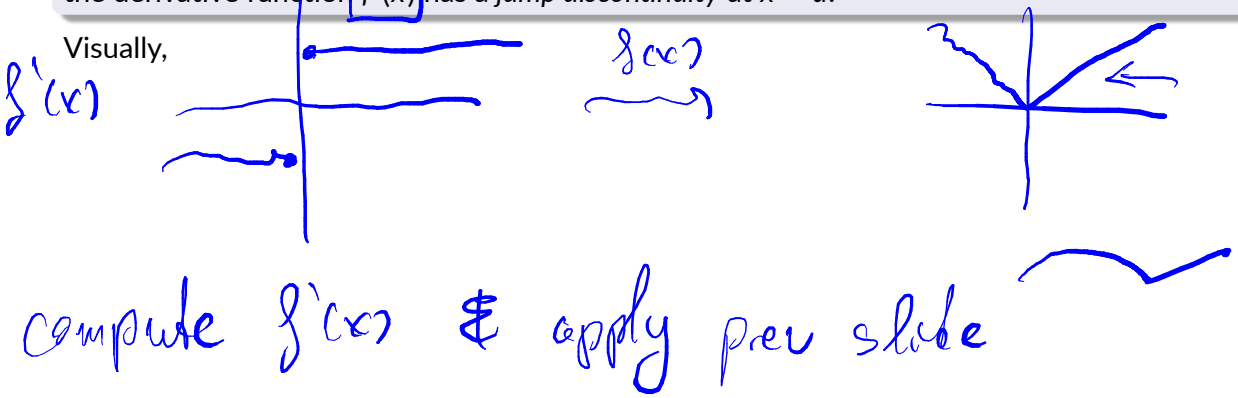
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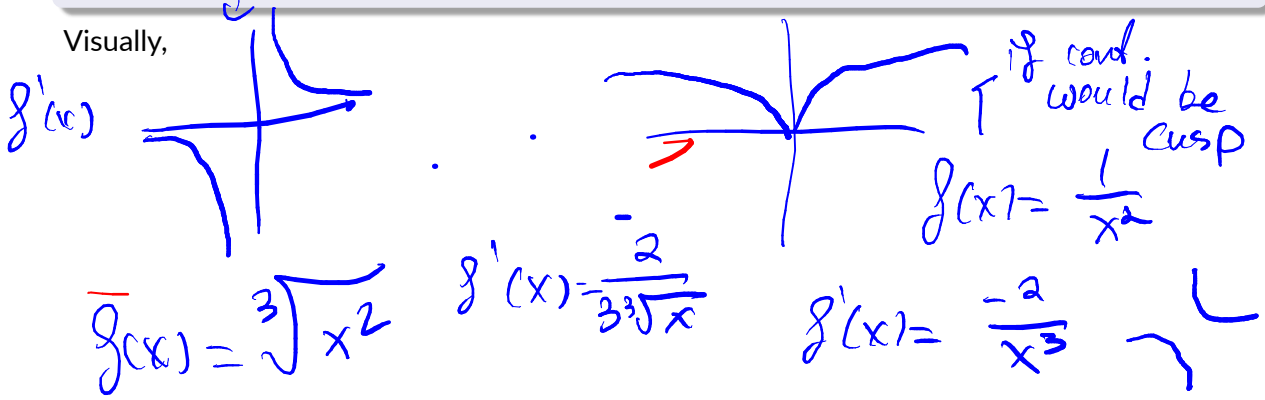
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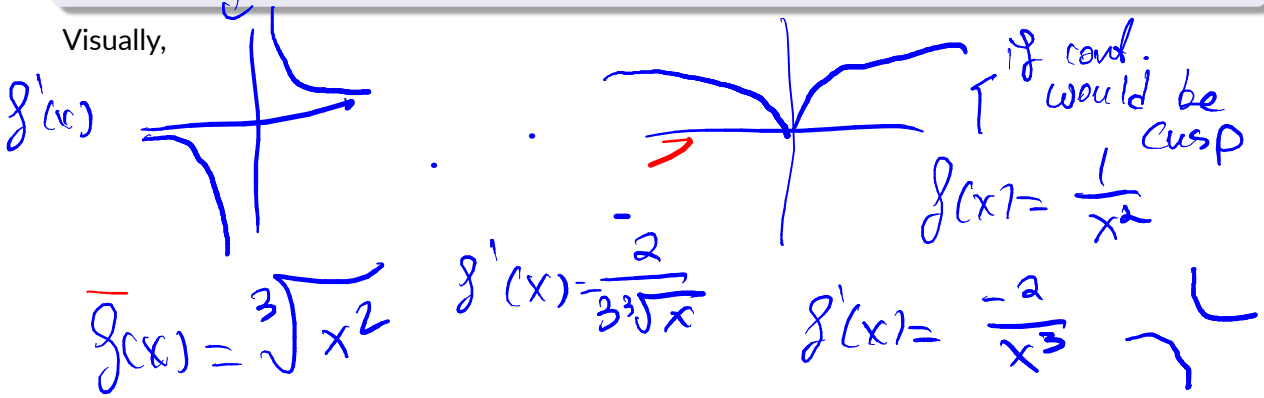
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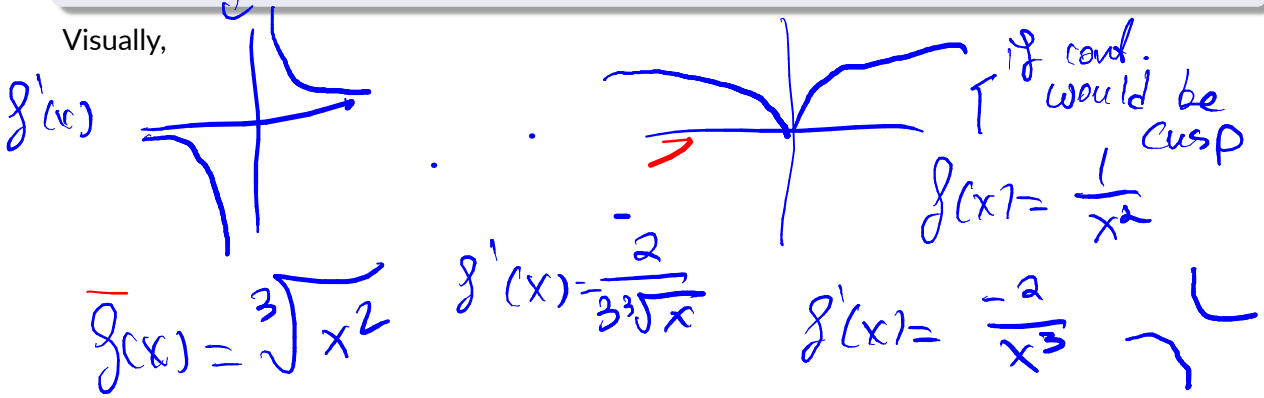
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Classifying Failures of Differentiability: Vertical Tangents

Vertical Tangent Singularity

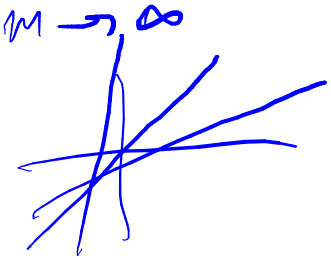
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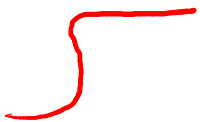
$$\lim_{x \to a} |f'(x)| = +\infty. \qquad \lim_{x \to a} f'(x) = \pm\infty \tag{5}$$



$f(x) = \sqrt[3]{x}$

$f'(x) = \frac{1}{3\sqrt[3]{x^2}}$

$\lim_{x \to 0} f'(x) = +\infty$





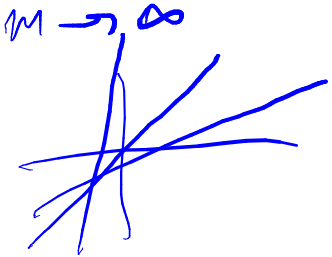
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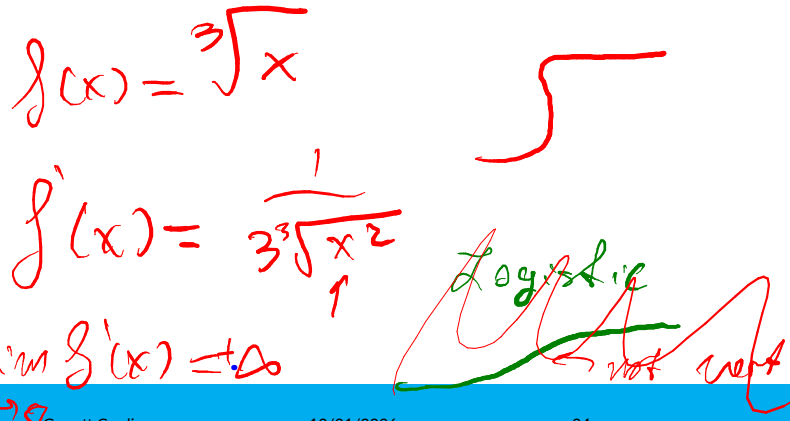
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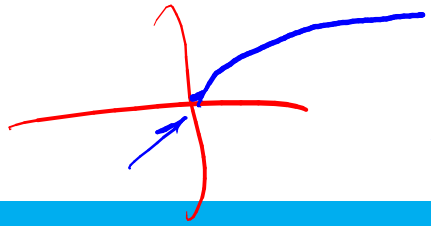
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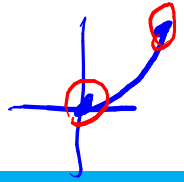
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$$f'(0) = \lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0} \frac{\sqrt{x}}{x}$$





x^2 $[0, 1)$

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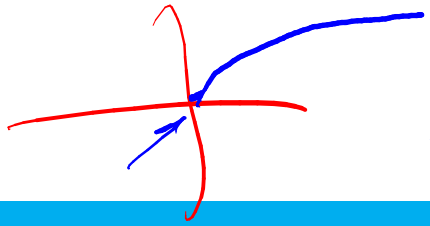
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Intuituion: Functions whose graphs are continuous, and corner/cusp/vertical tangent free.

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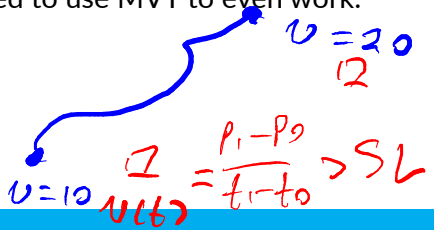
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3. Lots of the really useful tools in multivariable calculus need to use MVT to even work.

"mixed partials"

"Fun" application: $x(t)$



Worksheet Time

Let's get into groups and work on the worksheet!

End of Class!

That’s all folks!